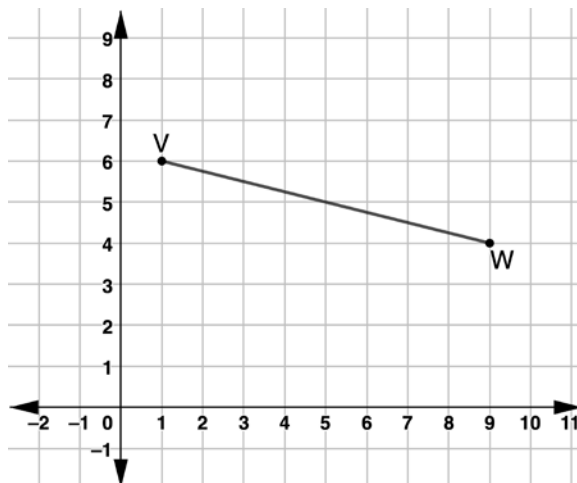


Geometry
Unit 13
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

Use the following information for questions 1-4. Point V has coordinates (1,6) and point W has coordinates (9,4).



1. Complete the table to calculate $\frac{1}{2}V + \frac{1}{2}W$.

	x-value	y-value
$W(1,6)$	$\frac{1}{2}(1)$	$\frac{1}{2}(6)$
$V(9,4)$	$\frac{1}{2}(9)$	$\frac{1}{2}(4)$
Total	5	5

2. What is the midpoint of line segment VW ?

$$\text{Midpoint} = \left(\frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2} \right) = \left(\frac{9+1}{2}, \frac{6+4}{2} \right) = \left(\frac{10}{2}, \frac{10}{2} \right) = (5,5)$$

3. Complete the expression that can be used to find the point that partitions the line segment in a 1:1 ratio.

$$\frac{1}{2}V + \frac{1}{2}W$$

4. Find the point that partitions directed line segment VW in a 2:1 ratio.

$$\frac{1}{3}V + \frac{2}{3}W = \left(\frac{1}{3}(1) + \frac{2}{3}(9), \frac{1}{3}(6) + \frac{2}{3}(4) \right) = \left(6\frac{1}{3}, 4\frac{2}{3} \right)$$

Use the following information for questions 5-6. Line segment FW has endpoints at $F(-10, -10)$ and $W(6, -2)$.

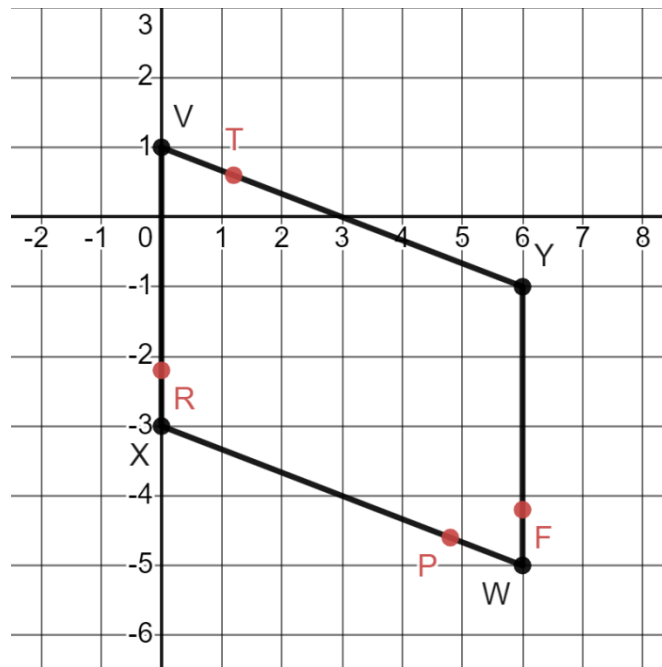
5. Write an expression for the point that partitions directed line segment FW in a ratio of 3 to 5.

$$\frac{5}{8}F + \frac{3}{8}W$$

6. Use the expression to find the point that partitions directed line segment FW in a 3:5 ratio.

$$\left(\frac{5}{8}(-10) + \frac{3}{8}(6), \frac{5}{8}(-10) + \frac{3}{8}(-2)\right) = (-4, -7)$$

Use the following information for questions 7-10. $VXWY$ has vertices at $V(0,1)$, $X(0,-3)$, $W(6,-5)$, and $Y(6,-1)$.



7. Find the point that partitions $\overline{WX}$ in a 1:4 ratio. Label it P .

$$\frac{4}{5}W + \frac{1}{5}X = \left(\frac{4}{5}(6) + \frac{1}{5}(0), \frac{4}{5}(-5) + \frac{1}{5}(-3)\right) = \left(4\frac{4}{5}, -4\frac{3}{5}\right) \text{ or } (4.8, -4.6)$$

8. Find the point that partitions $\overline{WY}$ in a 1:4 ratio. Label it F .

$$\frac{4}{5}W + \frac{1}{5}Y = \left(\frac{4}{5}(6) + \frac{1}{5}(6), \frac{4}{5}(-5) + \frac{1}{5}(-1)\right) = (6, -4.2)$$

9. Find the point that partitions $\overline{XV}$ in a 1:4 ratio. Label it R .

$$\frac{4}{5}X + \frac{1}{5}V = \left(\frac{4}{5}(0) + \frac{1}{5}(0), \frac{4}{5}(-3) + \frac{1}{5}(1)\right) = (0, -2.2)$$

10. Find the point that partitions $\overline{VY}$ in a 1:4 ratio. Label it T .

$$\frac{4}{5}V + \frac{1}{5}Y = \left(\frac{4}{5}(0) + \frac{1}{5}(6), \frac{4}{5}(1) + \frac{1}{5}(-1)\right) = (1.2, 0.6)$$